

Toward Zero Waste Rivers

CleanRiver Bot is an autonomous eco-friendly robot boat designed to collect plastic waste from rivers, lakes, and coastal waters.

Project Vision

Our mission is to restore cleaner waterways using intelligent robotics and renewable energy systems. By combining innovation, sustainability, and automation, CleanRiver Bot helps reduce water pollution and protect marine ecosystems.

How CleanRiver Bot Works

The robot moves autonomously across the water surface using propellers and smart navigation systems. Ultrasonic sensors detect obstacles and help the robot avoid collisions. A front conveyor and collection mechanism gather floating plastic waste into an internal storage container.

Solar Powered Technology

A compact solar panel mounted on top of the robot helps recharge the battery system, reducing energy consumption and enabling longer operation in outdoor environments.

Key Features

- Autonomous navigation
- Plastic collection system
- Solar-powered charging
- Ultrasonic obstacle detection
- OLED display and LED indicators
- Bluetooth or mobile control support
- Waterproof robotic design

Environmental Impact

CleanRiver Bot supports environmental protection by reducing plastic pollution before waste reaches the oceans. The project raises awareness about sustainability while demonstrating the future of eco robotics.

Future Goals

Our long-term goal is to deploy smarter and larger versions of CleanRiver Bot in major river systems around the world by 2030. Future upgrades may include AI waste detection, GPS mapping, and real-time environmental monitoring.

CleanRiver Bot — Technology for a Cleaner Future